Cortex-M4 Interrupt Priorities

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ARMv7-M Architecture Ref Manual

RM0432 STM32L4+ Technical reference manual

PM0214 STM32 Cortex-M4 Programming Manual

Figures from "The Definitive Guide to ARM® Cortex®-M3 and Cortex®-M4 Processors" by Joseph Yiu

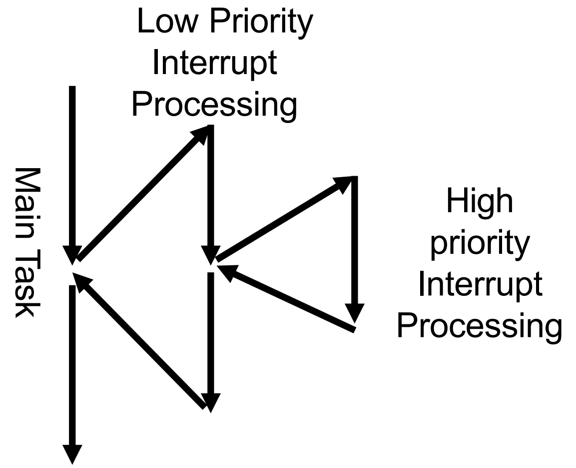
Acknowledgement: Slides adapted from Dr Rajesh Panicker

OVERVIEW

- Priority Levels: Exceptions and Interrupts
- Priority Types: Preempt and Sub-priority
- Configuration of Priorities
    - Interrupt Priority Register or Priority Level Register
    - Priority Group Setting

Priority Levels of Exceptions and Interrupts

- A higher priority (smaller priority level) exception/ interrupt can pre-empt a lower priority one
- This is called nested interrupts
- 3 exceptions have fixed priority levels
  - Reset (-3); NMI (-2) and hard fault (-1)
  - Negative levels is to assure the highest priorities
- Other exceptions and interrupts have programmable (user configurable) priority levels
  - Up to 256 levels (up to 128 levels of preemption), exact number of levels depends on the specific SoC
  - Exception number is used as a tie-breaker (a sub-sub-priority) if two interrupts have identical priority settings



Cortex-M4 Exceptions and Interrupts

Exception Number	Exception Type	Priority	Descriptions
1	Reset	–3 (Highest)	Reset
2	NMI	–2	Non-Maskable Interrupt (NMI), can be generated from on chip peripherals or from external sources.
3	Hard Fault	–1	All fault conditions, if the corresponding fault handler is not enabled
4	MemManage Fault	Programmable	Memory management fault; MPU violation or program execution from address locations with XN (eXecute Never) memory attribute.
5	Bus Fault	Programmable	Bus error; usually occurs when AHB interface receives an error response from a bus slave (also called *prefetch abort* if it is an instruction fetch or *data abort* if it is a data access). Can also be caused by other illegal accesses.
6	Usage Fault	Programmable	Exceptions due to program error or trying to access co-processor (the Cortex-M3 and Cortex-M4 processor do not support co-processors).
7–10	Reserved	NA	–
11	SVC	Programmable	SuperVisor Call; usually used in OS environment to allow application tasks to access system services.
12	Debug Monitor	Programmable	Debug monitor; exception for debug events like breakpoints, watchpoints when software based debug solution is used.
13	Reserved	NA	–
14	PendSV	Programmable	Pendable service call; An exception usually used by an OS in processes like context switching.
15	SYSTICK	Programmable	System Tick Timer; Exception generates by a timer peripheral which is included in the processor. This can be used by an OS or can be used as a simple timer peripheral.

Exception Number	Exception Type	Priority	Descriptions
16	Interrupt #0	Programmable	It can be generated from on chip peripherals or from external sources.
17	Interrupt #1	Programmable	
…	…	…	
255	Interrupt #239	Programmable	

Priority Types: Preempt priority and Sub-priority

- Preempt priority: Defines whether an exc./int. handler can execute when CPU is running another exc./int. handler
  - Interrupts with significant difference in urgency / importance = diff. preempt priorities
- Sub-priority: Is used when 2 exceptions/interrupts with the same preempt priority have occurred (and are pending). Higher sub-priority one will be handled first
  - Interrupts with only slight difference in urgency; not worth the preemption overhead = same preempt priorities, diff. sub-priorities

Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0
Implemented			Not implemented				

Priority Level Register with 3 bits implemented

Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0
Implemented				Not implemented			

Priority Level Register with 4 bits implemented

PRIORITY TYPES: PREEMPT PRIORITY AND SUB-PRIORITY

- Each exc./int. has an Interrupt Priority Register (IPR) (or Priority Level Register) to store its preempt (left bits) and sub-priority (right bits) values.
- IPR has 8 bits but only some of the more significant bits are implemented. This number is called the IPR width, and depends on the specific SoC
  - The split of implemented (available) bits between preempt and subpriority is determined by PRIGROUP

Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0
Implemented			Not implemented				

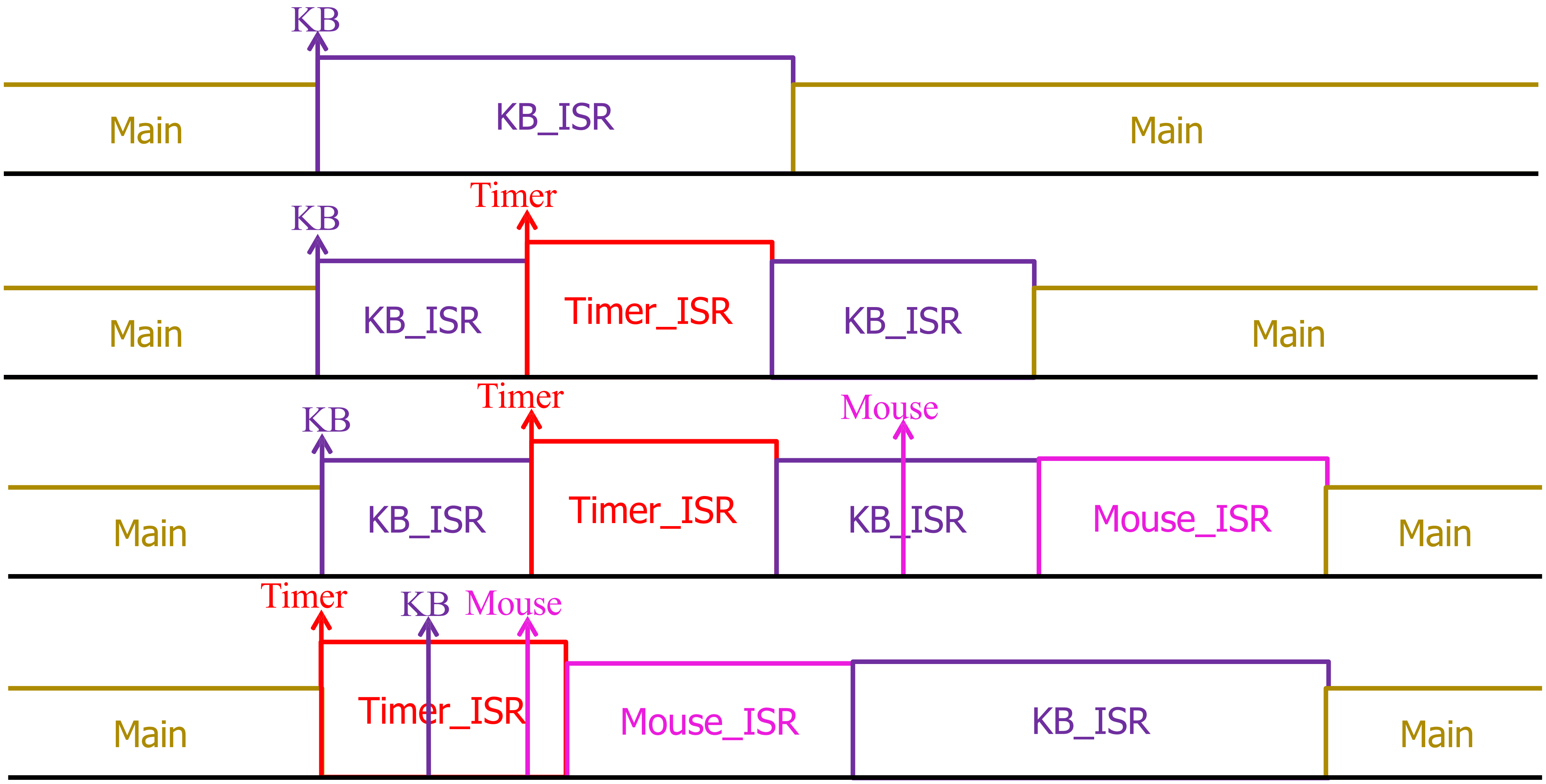
Priority Level Register with 3 bits implemented

Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0
Implemented				Not implemented			

Priority Level Register with 4 bits implemented

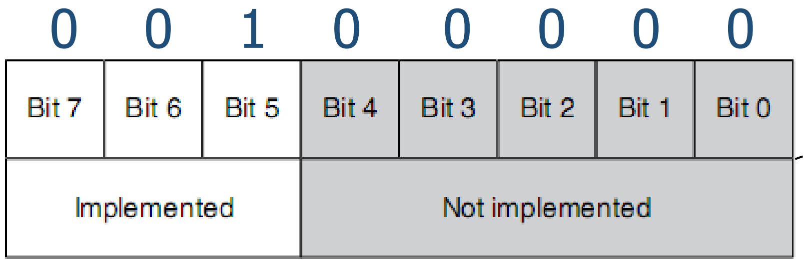
```
stm32l4s5xx.h:
#define __NVIC_PRIO_BITS 4
// This is not a 'setting' and should not be modified
```

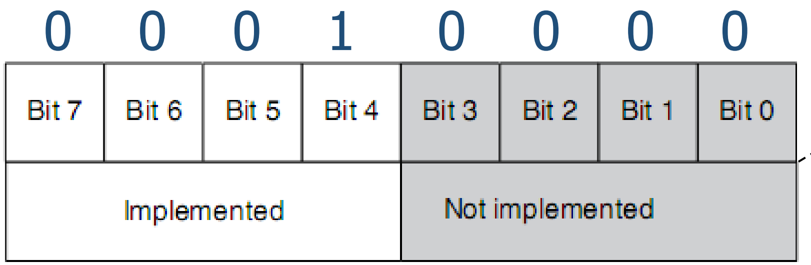
PRIORITY TYPES: PREEMPT PRIORITY AND SUB-PRIORITY

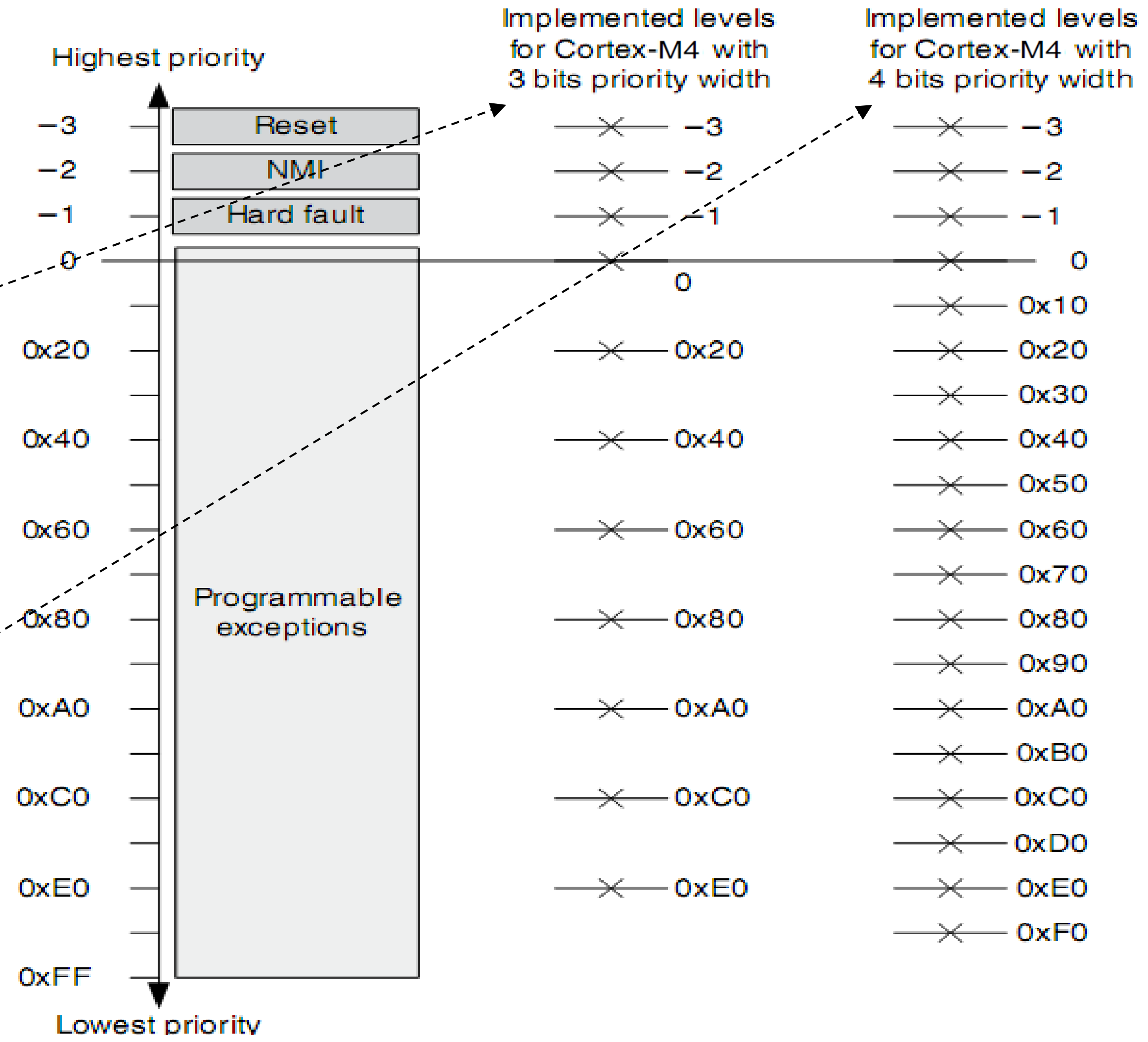


Pre(Timer) > Pre(KB) = Pre(Mouse)   Sub(Mouse) > Sub(KB)

Interrupt Priority Register







Bits not implemented are read as 0
Writes to them will be ignored

Priority group & 2 priority fields in Interrupt Priority Register

- In IPR, the division of preempt and sub-priority field is decided by a PRIGROUP (Priority group) in Application Interrupt and Reset Control Register (AIRCR) in System Control Block (SCB)
  - Number of bits implemented in the Interrupt Priority Register and the Priority group setting affects the effective preempt and sub-priority levels
  - PRIGROUP to be set is decided by the programmer based on the number of distinctly different priority levels required in the application

Table 7.4 Application Interrupt and Reset Control Register (Address 0xE000ED0C)

Bits	Name	Type	Reset Value	Description
31:16	VECTKEY	R/W	—	Access key; 0x05FA must be written to this field to write to this register, otherwise the write will be ignored; the read-back value of the upper half word is 0xFA05
15	ENDIANNESS	R	—	Indicates endianness for data: 1 for big endian (BE8) and 0 for little endian; this can only change after a reset
10:8	PRIGROUP	R/W	0	Priority group
2	SYSRESETREQ	W	—	Requests chip control logic to generate a reset
1	VECTCLRACTIVE	W	—	Clears all active state information for exceptions; typically used in debug or OS to allow system to recover from system error (Reset is safer)
0	VECTRESET	W	—	Resets the Cortex-M3 processor (except debug logic), but this will not reset circuits outside the processor

Table 7.5 Definition of Preempt Priority Field and Subpriority Field in a Priority Level Register in Different Priority Group Settings

Priority Group	Preempt Priority Field	Subpriority Field
0	Bit [7:1]	Bit [0]
1	Bit [7:2]	Bit [1:0]
2	Bit [7:3]	Bit [2:0]
3	Bit [7:4]	Bit [3:0]
4	Bit [7:5]	Bit [4:0]
5	Bit [7:6]	Bit [5:0]
6	Bit [7]	Bit [6:0]
7	None	Bit [7:0]

PRIORITY FIELDS IN A 3-BIT IPR WITH PRIGROUP SET TO 5

IPR width = 3

PRIGROUP = 5

Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0
Implemented			Not implemented				

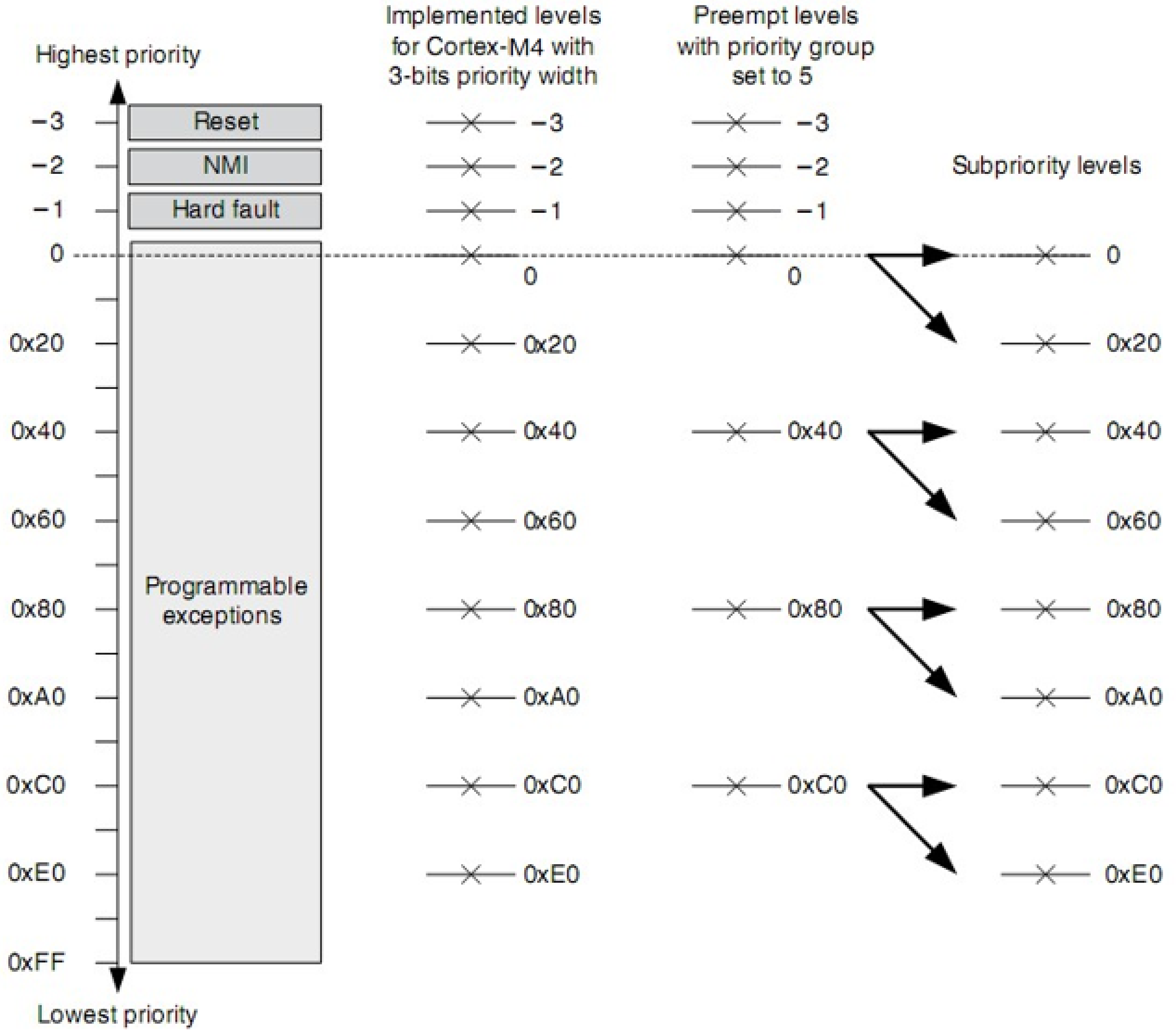
Table 7.5 Definition of Preempt Priority Field and Subpriority Field in a Priority Level Register in Different Priority Group Settings

Priority Group	Preempt Priority Field	Subpriority Field
0	Bit [7:1]	Bit [0]
1	Bit [7:2]	Bit [1:0]
2	Bit [7:3]	Bit [2:0]
3	Bit [7:4]	Bit [3:0]
4	Bit [7:5]	Bit [4:0]
5	Bit [7:6]	Bit [5:0]
6	Bit [7]	Bit [6:0]
7	None	Bit [7:0]

Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0
Preempt priority		Sub-priority	Not implemented				

Definition of Priority Fields in a 3-bit Priority Level Register with Priority Group Set to 5

PRIORITY FIELDS IN A 3-BIT IPR WITH PRIGROUP SET TO 5



Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0
Preempt priority		Sub-priority	Not implemented				
0	0	0	0	0	0	0	0
0	0	1	0	0	0	0	0
0	1	0	0	0	0	0	0
0	1	1	0	0	0	0	0
1	0	0	0	0	0	0	0
1	0	1	0	0	0	0	0
1	1	0	0	0	0	0	0
1	1	1	0	0	0	0	0

PRIORITY FIELDS IN A 3-BIT IPR WITH PRIGROUP SET TO 1

IPR width = 3

PRIGROUP = 1

Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0
Implemented			Not implemented				

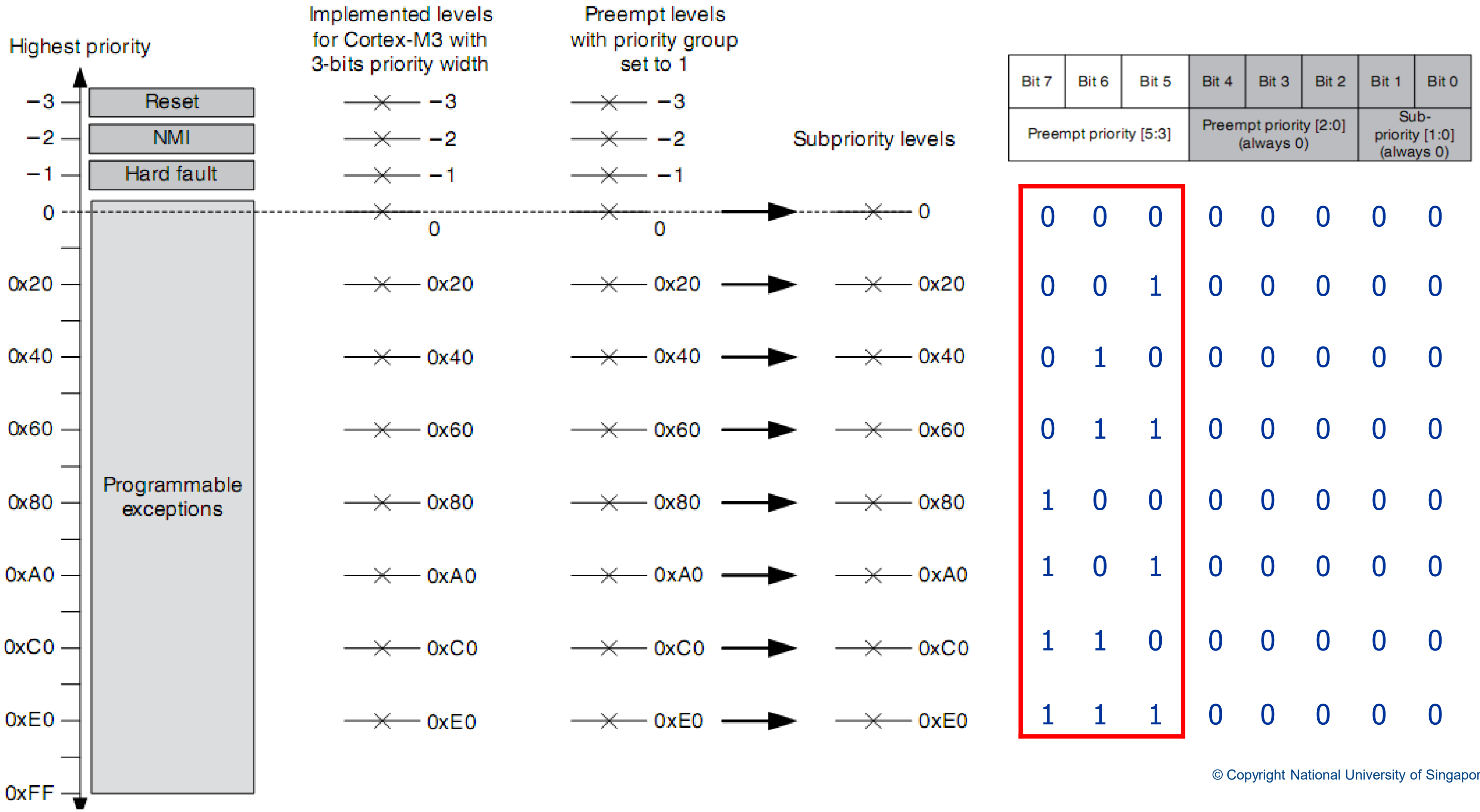
Table 7.5 Definition of Preempt Priority Field and Subpriority Field in a Priority Level Register in Different Priority Group Settings

Priority Group	Preempt Priority Field	Subpriority Field
0	Bit [7:1]	Bit [0]
1	Bit [7:2]	Bit [1:0]
2	Bit [7:3]	Bit [2:0]
3	Bit [7:4]	Bit [3:0]
4	Bit [7:5]	Bit [4:0]
5	Bit [7:6]	Bit [5:0]
6	Bit [7]	Bit [6:0]
7	None	Bit [7:0]

Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0
Preempt priority [5:3]			Preempt priority [2:0] (always 0)			Sub-priority [1:0] (always 0)	

Definition of Priority Fields in a 3-bit Priority Level Register with Priority Group Set to 1

PRIORITY FIELDS IN A 3-BIT IPR WITH PRIGROUP SET TO 1



CONCLUSION

- Priority Levels: Exceptions and Interrupts
  - Fixed priority levels
  - Configurable priority levels
  - Exception Number
- Priority Types
  - Preempt priority
  - Sub-priority
- Interrupt Priority Register (IPR)
  - SoC specific IPR width
- Priority Group Settings in Application Interrupt and Reset Control Register (AIRCR)